nature portfolio

Corresponding author(s): Rampi Ramprasad
Last updated by author(s): 12-08-2026

Machine Learning Checklist V 1.1

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended to provide structure for consistency and transparency in reporting of works using or developing Machine Learning models. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

1. Availability and reproducibility of Code and Data

Please select all that apply regarding the availability of the data and code used in the study.

☐ Code will be included in a CodeOcean capsule.

☑ The **source code** is included in the submission or available in a public repository:
Submitted supplementary software file Code_package.zip; public repository upon publication.

☐ A **compiled standalone version** of the software is included in the submission or available in a public repository:

☐ A **test dataset** and instructions/scripts for replicating the results are included in the submission or available in a public repository:

☑ A **Readme file** with instructions for installing and running the code is included in the submission or available in a public repository:
A README file and dependency information are included in the supplementary software package.

☑ The code is made available to reviewers during review.

☐ **Pretrained models** are used in the study and accessible through:

☑ **Pretrained models** are used in the study and are not accessible.

☑ The paper contains information on how to obtain code and data after publication.

2. Datasets

A. All data sources are listed in the paper.
   ☑ Yes
   ☐ No

B. The train, test and validation datasets are publicly available, and links/accession numbers have been provided in the manuscript or supplementary materials.
   ☐ Yes
   ☑ No

C. We have reported and discussed potential dataset biases in the paper. Where applicable, appropriate mitigation strategies were used.

☑ Yes dataset sources, labeling criteria, class imbalance, literature bias toward successful ROMP examples, and the construction of a diverse inactive set are described in the Methods section and Su

☐ No Explain why this information is not reported/not relevant

D. The data cleaning and preprocessing steps are clearly and fully described, either in text or as a code pipeline.

☑ Yes n, molecular representation construction, BFF descriptor calculation, and standardized recipe preparation.

☐ No Explain why this information is not reported/not relevant

E. Instances of combining data from multiple sources are clearly identified, and potential issues mitigated.

☑ Yes methyl-substituted derivatives, and prospective experimental data is described in the Methods and Supporting Information Sections S1.1–S1.3, S3, S4, and S9.

☐ No Explain why this information is not reported/not relevant

3. Model and training

A. What model architecture is the current model based on? GPT-3.5 and LoRA/QLoRA-fine-tuned LLaMA-3-8B models, Gaussian process classifiers

B. A Model Card is provided[1].

☐ Yes

☑ No

C. The model clearly splits data into different sets for training (model selection), validation (hyperparameter optimization), and testing (final evaluation).

☑ Yes

☐ No

D. The method of data splitting (e.g. random, cluster- or time-based splitting, forward cross-validation) is clearly stated.

☑ Yes Data splitting is described in the Methods and Supporting Information Sections S1.5, S1.5, and S1.6. ROMP feasibility models used stratified train/test splits, and the synthesis-procedure gene

☐ No Explain why this information is not reported/not relevant

E. The data splitting mimics anticipated real-world applications.

☑ Yes dures further test the intended use case: applying the workflow to new monomers not used in training.

☐ No Explain why this information is not reported/not relevant

F. The data splitting procedure has been chosen to avoid data leakage.

☑ Yes ords were consolidated by retaining the most complete primary entry, and prospective validation monomers were not used for training.

☐ No Explain why this information is not reported/not relevant

[1] https://huggingface.co/docs/hub/model-cards

G. The interpretability of the model has been studied and clearly validated.

☑ Yes 35, and 36. SHAP trends were compared with ROMP-relevant descriptors, and the heuristic explan er was evaluated against manually annotated incompatibility labels.

☐ No Explain why this information is not reported/not relevant

4. Evaluation

A. The performance metrics used are described and justified in the paper.

☑ Yes for ROMP feasibility prediction, R2 and RMSE for surrogate DFT descriptor prediction, and slot-level matching plus blinded expert ratings for procedure generation.

☐ No Explain why this information is not reported/not relevant

B. Cross-validation of the results is included.

☑ Yes

☐ No

C. Community-accepted benchmark datasets/tasks are used for comparisons.

☐ Yes State where this information can be found in the text

☑ No No community benchmark currently exists for ROMP feasibility prediction or ROMP procedure generation. The models were therefore compared on a curated ROMP dataset and further tested using external enyne and prospective experimental validation sets.

D. Baseline comparisons to simple/trivial models (for example, 1-nearest neighbour, random forest, most frequent class) are provided.

☑ Yes The study compares several chemically meaningful model families and molecular representations, including RC_CDC, polyBERT_CDC, DFTactual_CDC, DFTpredicted_CDC, CRT, and LLaMA

☐ No Explain why this information is not reported/not relevant

E. Benchmarks with current state-of-the-art are provided.

☐ Yes State where this information can be found in the text

☑ No No established state-of-the-art benchmark exists for this task. The study benchmarks internally across contemporary model types and molecular representations.

F. Ablation experiments are included.

☑ Yes tion, surrogate descriptor prediction, SHAP analysis, and slot-level evaluation of generated procedures.

☐ No Explain why this information is not reported/not relevant

G. The model has been tested on a fully independent dataset.

☑ Yes

☐ No

5. Computational resources

A. The paper contains information on hardware/computing resources that were used.

☐ Yes

☑ No

B. The paper includes information on the computational costs in terms of computation time, parallelization or carbon footprints estimates.

☐ Yes

☑ No